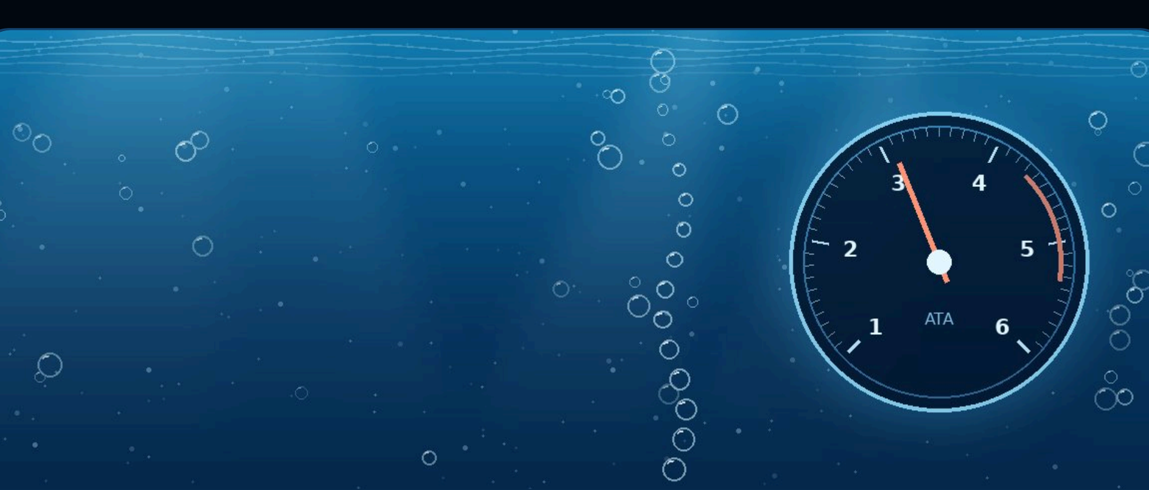


GLI AMICI DI BOYLE · NEWSLETTER

Boyle's Barometer

No. 1 · October 2026 · the pressure of the month, from the surface to the seabed

In this issue: who the Boyle Network is · upcoming events in Genoa, Varese and Bolzano · five readings from 2026 and a case report of ours · Boyle Bibliography and Diving-Boyle · the latest episodes of Sotto Pressione.

▼ −10 m · 2.0 ATA · The Network

THE BOYLE NETWORK

Extreme environment expertise, in a single network

Dear friends of Boyle,

with this first issue we are launching a regular update on the Boyle Network and on what happens within it. You are receiving the English edition: the Italian one goes out at the same time.

We come from all over Italy, from Turin to Palermo, from Venice to Sassari, by way of Milan, Varese, Como, Padua, Bologna, Perugia, Rome and Taranto, and from beyond its borders: **Switzerland, San Marino, Germany, Portugal, Brazil, Hong Kong and New Zealand**. In recent months the Network has grown to include polar and expedition medicine, space medicine, military and special operations medicine, medical intelligence and engineering applied to healthcare.

The strength of the Network lies in the range of professions it brings together:

Critical care and resuscitation
Anaesthesiologists and intensivists, from ECMO to trauma, from helicopter rescue to mountain, polar and expedition medicine.

Diving and hyperbaric medicine
Diving and hyperbaric physicians, hyperbaric technicians and nurses, an Italian Navy diver, instructors and explorers of wrecks and caves.

Specialties that meet the water
Cardiologists, endocrinologists and diabetologists working on fitness to dive and diving-related disease, a psychologist specialised in diving psychology.

Operational and military
Military and special operations medicine, TCCC, medical intelligence and CBRNE, search and rescue, mass gathering and emergency surgery.

Space
An analog astronaut, aerospace medicine education, astrobiology, space medicine.

Engineering and research
Biomedical, aerospace and electronic engineers, neuroscience, physiology, drones and artificial intelligence applied to healthcare.

Get to know each other, and let us know what you do

The Network works when people talk to each other. Every member has a personal profile on the Network page: **use it to get in touch with one another**, for a study, a congress, a consultation, a course. And if you are running an initiative, write to us: **we will make it known to the whole Network**, on the website and here in the newsletter.

One small step helps everyone: log in with your code, click the pencil icon and update your bio, publications and links. The more complete your profile, the easier you are to find.

Explore the Network | Write to us

▼ −20 m · 3.0 ATA · Events

UPCOMING EVENTS

Where to find us

IL LIMITE È
LA CONOSCENZA
DECOMPRESSIONE, PREVENZIONE
E SOCCORSO IN IMMERSIONE

SATURDAY 7 NOVEMBER · GENOA · FREE ENTRY

Il limite è la conoscenza (The limit is knowledge)

FIAS conference on decompression, prevention and rescue in diving, held in Italian. «Luigi Durand De La Penne» Auditorium, ITTL Nautico San Giorgio, Porto Antico, from 3 pm. Talks by Silvio Zerbi, Andrea Murdock Alpini, Marco Mori and Giovanni Cappa. Registration required by 2 November.

Register →

FRIDAY 27 NOVEMBER · VARESE · 4 ECM CREDITS

Diving and Hyperbaric Medicine: from physiology to clinical practice

Aula Magna, Collegio Cattaneo, 1:45–6:30 pm, in Italian. Chaired by Silvio Zerbi and Paolo Severgnini; speakers include Vincenzo Benenati, Claudio Camponovo (DAN Europe), Carmela Graci, Andrea Moriondo, Rosario Cacopardo and Andrea Murdock Alpini. Free, up to 90 participants, registration on the training portal (event ID 223241.1).

Register →

Hypercare
Emergency

21–22 JANUARY 2027 · BOLZANO · SAVE THE DATE

Hypercare · Emergency Medicine Course

With ITECC and Eurac Research at terraXcube: burns, hypothermia and avalanches, decompression illness, from pathophysiology to treatment. Full programme and registration coming soon: we will share them as soon as they are final.

Poster →

Just concluded

Inside Trauma Care · ON FIELD, 10 September in Bresso: a day of major incident scenario simulation. Thanks to our partner ITECC, chaired by Network member Erika Borotto, and to everyone who took part.

In recent months the Network has also shared the **50th EUBS Congress** in Geneva, the second edition of the **Diving Psychology Course** on the island of Elba, proposed by Lidia Porri, and the **Luna e Medicina** (Moon and Medicine) conference in Duno, proposed by Francesco Baggio. This is exactly how we want to work.

Organising an event? Let us know

Courses, congresses and training days on extreme environments, diving and hyperbaric medicine: we publish them on the website and, from this issue, in the newsletter too. We do this voluntarily and free of charge, to help quality initiatives reach the people who will truly value them.

Propose an event

▼ −30 m · 4.0 ATA · Readings

READINGS FROM 2026

Five papers not to miss, plus one of ours

The most relevant and most intriguing papers published this year, with our comments, ordered by the standing of the journal. Ours comes last, out of modesty.

1 **CA: A Cancer Journal for Clinicians**
HYPERBARIC · ONCOLOGY · CLINICAL REVIEW

Hyperbaric oxygen therapy for chronic radiotherapy-related adverse effects: a clinically focused review

Dejonckheere et al. · 2026 · doi:10.3322/caac.70058

One of the most widely read oncology journals in the world devotes a clinical review to hyperbaric oxygen therapy for late radiation injury: skin fibrosis, osteoradionecrosis, radiation cystitis and proctitis. The authors review randomised trials, cohorts and real-world data and point out the open gaps: patient selection, optimal protocols, long-term outcomes. The signal is clear: hyperbaric medicine is entering the language of care for long-term cancer survivors.

2 **Progress in Retinal and Eye Research**
SPACE MEDICINE · REVIEW

Ocular changes in human spaceflight

Gupta et al. · 2026 · doi:10.1016/j.preteyeres.2026.101535

An up-to-date synthesis of how spaceflight changes the eye: spaceflight-associated neuro-ocular syndrome (SANS), corneal epithelial injury, possible radiation-related lens changes, retinal and choroidal changes. Most interesting is the section on the limits of diagnosis and treatment in microgravity, from imaging to intracranial pressure measurement: the theme of operational constraints that we know well underwater too.

3 **Journal of Crohn's and Colitis**
HYPERBARIC · GASTROENTEROLOGY · PROSPECTIVE PILOT TRIAL

Hyperbaric oxygen therapy alters bowel perfusion and improves outcomes in patients with treatment-refractory ulcerative colitis: a prospective pilot trial

Mulders et al. · 2026 · doi:10.1093/ecco-jcc/jjag065

Sixteen patients with ulcerative colitis refractory to at least two advanced therapies, treated with 10 or 20 sessions at 2.4 ATA. At twelve weeks, clinical and endoscopic response was seen in 2 of 8 patients in the 10-session group and 4 of 8 in the 20-session group, and contrast-enhanced ultrasound showed increased bowel perfusion in responders. Small numbers and an open-label design, but a pathophysiological rationale worth following closely.

4 **Diving and Hyperbaric Medicine**
TECHNICAL DIVING · REBREATHER · OBSERVATIONAL STUDY

The effect of hyperoxic environment on lung function and myopisation in an extended closed-circuit rebreather diving expedition

Sokolowski-Wojnicki et al. · 2026;56(3):294-302 · doi:10.28920/dhm56.3.294-302

Thirty-one days in Chuuk Lagoon, twelve divers, 173 closed-circuit rebreather dives, a mean maximum depth of 38 metres and almost a hundred minutes per dive. Spirometry and visual acuity checked weekly: no significant change overall, but one diver lost lung function and another developed mild obstruction. Within oxygen exposure limits, long expeditions are sustainable, but individual variability is real.

5 **Diving and Hyperbaric Medicine**
FREEDIVING · DECOMPRESSION ILLNESS · CASE REPORT

In-water recompression for severe decompression illness in competitive freediving: a case report

Johnston et al. · 2026;56(3):384-387 · doi:10.28920/dhm56.3.384-387

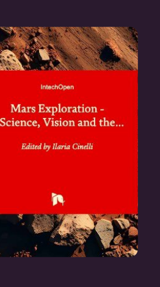
A competitive freediver surfaces from 117 metres with severe neurological symptoms: hemiparesis, aphasia, altered consciousness. With the hyperbaric chamber 24–48 hours away, the team attempts in-water recompression on pure oxygen with a full-face mask, starting at 18 metres: substantial recovery, and complete resolution after a second treatment. The authors themselves stress the risks, oxygen toxicity and drowning, and the safety issues that emerged. A case worth knowing, not a protocol to imitate.

6 **Diving and Hyperbaric Medicine** · FROM THE NETWORK
CARDIOLOGY · DIVING · HYPERBARIC · CASE REPORT

Returning to diving and hyperbaric exposure after pulmonary vein isolation for atrial fibrillation

Zerbi S, Tarsia L, Benenati V, Nicosia D, Bosco G, Paganini M · 2026;56(2):203-207 · doi:10.28920/dhm56.2.203-207

Our contribution. An experienced 46-year-old diver who had undergone successful pulmonary vein isolation was monitored during a 42-metre dive with a marinised 12-lead Holter and during a hyperbaric chamber compression to 2.8 ATA, with his implantable loop recorder active throughout. No atrial fibrillation recurrence, no arrhythmias, stable heart rate; the loop recorder remained fully functional after both exposures, while telemetry failed because of shielding by the chamber's steel walls. It is the first hyperbaric electrophysiology assessment described in a post-ablation patient: a starting point for fitness-to-dive decisions in these patients.



FROM THE NETWORK

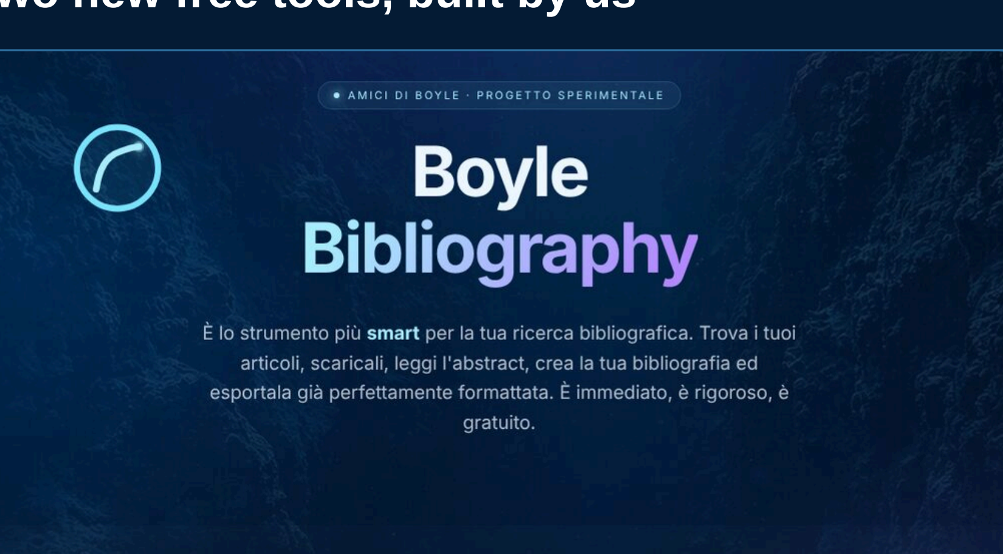
First Cut on Mars, an open access chapter on IntechOpen by Zerbi, Tarsia, Baggio and Giacon: what surgery can realistically support an outpost on Mars, with limited resources, delayed communication with Earth and a team that has to be self-sufficient.

Read the chapter →

▼ −40 m · 5.0 ATA · Our tools

NEW ON THE WEBSITE

Two new free tools, built by us



È lo strumento più **smart** per la tua ricerca bibliografica. Trova i tuoi articoli, scaricali, leggi l'abstract, crea la tua bibliografia ed esportala già perfettamente formattata. È immediato, è rigoroso, è gratuito.

BOYLE BIBLIOGRAPHY

More than PubMed, on a single page

It searches PubMed and OpenAlex at the same time and hands you a ready-made bibliography. Everything that usually takes three websites and a reference manager:

- ✓ **Search in English or Italian**: the most common field terms are translated automatically.
- ✓ **Two databases together**, PubMed and OpenAlex, with no duplicates.
- ✓ **Authors with the same name kept apart** thanks to PubMed's official tag.
- ✓ **Real filters** by study type (review, RCT, meta-analysis, guideline) and year.
- ✓ **Read the abstract instantly**, without opening anything, and download open access PDFs straight away.
- ✓ **Vancouver bibliography** ready to paste, or BibTeX, RIS or Word for Zotero, Mendeley and EndNote.
- ✓ **Real data only**: no entry written by artificial intelligence, no invented sources.

Try Boyle Bibliography

ESPLORA PER REGIONE

Ecco i diving italiani. Qui si impara ad esplorare la parte nascosta del mondo.

Trova i migliori diving italiani.

DIVING-BOYLE

Where Italy dives

The map of Italian dive centres curated by Gli Amici di Boyle: **over 700 centres in 20 regions**. Search by name or town and the listing opens straight away, or explore region by region. Useful if you are planning a dive trip to Italy or need to recommend a centre to a patient.

The data come from public sources and we are verifying them: if you know or run a centre, every listing has a form to report corrections.

Explore the map

▼ −50 m · 6.0 ATA · Sotto Pressione

SOTTO PRESSIONE · THE PODCAST

SOTTO PRESSIONE
di Luigi Tarsia

The latest episodes of Luigi Tarsia's podcast, in Italian:

- ▶ **Le leggi dei gas: Boyle, Charles e Gay-Lussac** · Ep. 13
- ▶ **HBOT e medicina spaziale** · Ep. 12
- ▶ **Questa non è un'esercitazione**, the air leak on the ISS

All episodes →

▼ 0 m · 1.0 ATA · Ascent

Have you read a paper that deserves space in the next issue, or do you have an initiative to share? Just reply to this email: this newsletter is yours too.

Thank you for being part of this Network. Until our next dive.

Gli Amici di Boyle
Research & Education

> *The sea, once it casts its spell, holds one in its net of wonder forever.*
>
> JACQUES-YVES COUSTEAU

OUR PARTNERS

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